An Analysis of Hierarchical clustering

Hierarchical clustering

Hierarchical clustering -- Part 1

Comparison of linkage methods The choice of linkage criterion significantly affects the shape and quality of the resulting clusters. Each method has distinct strengths and weaknesses depending on the structure of the data. Single linkage (also called the nearest-neighbor method) defines the distance between two clusters as the minimum distance between any pair of points, one from each cluster. This method can detect and handle clusters of arbitrary shape, including elongated and non-globular structures. However, it is highly sensitive to noise and outliers, because a single pair of close points is sufficient to merge two clusters. Complete linkage defines the distance between two clusters as the maximum distance between any pair of points across the two clusters. It is less susceptible to noise and outliers than single linkage. However, complete linkage is biased toward producing globular clusters of similar size and may incorrectly break apart large or irregularly shaped clusters. Average linkage (specifically UPGMA, the unweighted pair group method with arithmetic mean) defines the distance between two clusters as the average of all pairwise distances between points in the two clusters. It represents a compromise between the extremes of single and complete linkage. Average linkage is more robust to noise than single linkage but, like complete linkage, it tends to be biased toward detecting globular clusters.

Hierarchical clustering -- Part 2

Pop i ∗ {\displaystyle i^{*}} from its old cluster C ∗ {\displaystyle C_{*}} and put it into a new splinter group C new = { i ∗ } {\displaystyle C_{\textrm {new}}=\{i^{*}\}} . As long as C ∗ {\displaystyle C_{*}} isn't empty, keep migrating objects from C ∗ {\displaystyle C_{*}} to add them to C new {\displaystyle C_{\textrm {new}}} . To choose which objects to migrate, don't just consider dissimilarity to C ∗ {\displaystyle C_{*}} , but also adjust for dissimilarity to the splinter group: let i ∗ = arg ■ max i ∈ C D (i) {\displaystyle i^{*}=\arg \max _{i\in C}D(i)} where we define D (i) = 1 | C ∗ | − 1 ∑ j ∈ C ∗ ■ { i } δ (i , j) − 1 | C new | ∑ j ∈ C new δ (i , j) {\displaystyle D(i)={\frac {1}{|C_{*}|-1}}\sum _{j\in C_{*}\setminus \{i\}}\delta (i,j)-{\frac {1}{|C_{\textrm {new}}|}}\sum _{j\in C_{\textrm {new}}}\delta (i,j)} , then either stop iterating when D (i ∗) < 0 {\displaystyle D(i^{*})<0} , or migrate i ∗ {\displaystyle i^{*}} . Add C new {\displaystyle C_{\textrm {new}}} to C {\displaystyle {\mathcal {C}}} . Intuitively, D (i) {\displaystyle D(i)} above measures how strongly an object wants to leave its current cluster, but it is attenuated when the object wouldn't fit in the splinter group either. Such objects will likely start their own splinter group eventually. The dendrogram of DIANA can be constructed by letting the splinter group C new {\displaystyle C_{\textrm {new}}} be a child of the hollowed-out cluster C ∗ {\displaystyle C_{*}} each time. This constructs a tree with C 0 {\displaystyle C_{0}} as its root and n {\displaystyle n} unique single-object clusters as its leaves.

Hierarchical clustering -- Part 3

Some of these can only be recomputed recursively (WPGMA, WPGMC), for many a recursive computation with Lance-Williams-equations is more efficient, while for other (Hausdorff, Medoid) the distances have to be computed with the slower full formula. Other linkage criteria include:

Hierarchical clustering -- Part 4

Selecting the number of clusters In hierarchical clustering, the number of clusters is not determined automatically and must be chosen by analyzing the dendrogram. A common approach is to “cut” the dendrogram at a selected height, which partitions the data into clusters based on where the cut intersects the tree structure. The choice of cut level has a significant impact on the resulting clustering. One widely used heuristic is to identify large vertical gaps in the dendrogram, which correspond to substantial increases in the distance between successive merges. Cutting the dendrogram at a height just before such a gap can produce clusters that reflect natural groupings in the data. This is based on the observation that merges occurring at much larger distances often combine dissimilar clusters.

Hierarchical clustering -- Part 5

This definition captures the largest distance from a point in one set to the closest point in the other set. Equivalently, it can be interpreted as the smallest radius r {\displaystyle r} such that every point of A {\displaystyle A} lies within distance r {\displaystyle r} of some point in B {\displaystyle B} , and every point of B {\displaystyle B} lies within distance r {\displaystyle r} of some point in A {\displaystyle A} . In this sense, the Hausdorff distance measures the maximum mismatch between two sets. In hierarchical clustering, Hausdorff linkage uses d H (A , B) {\displaystyle d_{H}(A,B)} as the dissimilarity measure between clusters. Initially, each data point forms its own cluster, and the Hausdorff distance between singleton sets reduces to the original metric. At each step, the two clusters with the smallest Hausdorff distance are merged, and distances are recomputed using the same set-based definition. This iterative process continues until all points belong to a single cluster, producing a hierarchical structure (dendrogram). Compared to other linkage criteria such as single, complete, or average linkage, Hausdorff linkage operates on a set-to-set basis rather than relying solely on pairwise point distances. As a result, it captures the global geometry of clusters and can better preserve structural features of the data. It has been shown to provide behavior intermediate between single and complete linkage, avoiding the chaining effect of the former while reducing the tendency of the latter to overestimate distances between clusters. However, these advantages come with increased computational cost. Computing the Hausdorff distance between two clusters typically requires evaluating distances between all pairs of points across the sets, often via a max–min procedure over a distance matrix. This makes Hausdorff linkage more computationally expensive than simpler linkage criteria, particularly for large datasets. Overall, Hausdorff linkage provides a mathematically well-founded clustering approach that is effective for distinguishing complex structures in data.

Hierarchical clustering -- Part 6

Complexity The standard algorithm for hierarchical agglomerative clustering (HAC) has a time complexity of O (n 3) {\displaystyle {\mathcal {O}}(n^{3})} and requires Ω (n 2) {\displaystyle \Omega (n^{2})} memory, which makes it too slow for even medium data sets. However, for some special cases, optimal efficient agglomerative methods (of complexity O (n 2) {\displaystyle {\mathcal {O}}(n^{2})}) are known: SLINK for single-linkage and CLINK for complete-linkage clustering. With a heap, the runtime of the general case can be reduced to O (n 2 log ■ n) {\displaystyle {\mathcal {O}}(n^{2}\log n)} instead of O (n 3) {\displaystyle {\mathcal {O}}(n^{3})} , at the cost of additional the memory requirements. In many cases, the memory overheads of this approach are too large to make it practically usable. Methods exist which use quadtrees that demonstrate O (n 2) {\displaystyle {\mathcal {O}}(n^{2})} total running time with O (n) {\displaystyle {\mathcal {O}}(n)} space. Divisive clustering with an exhaustive search is O (2 n) {\displaystyle {\mathcal {O}}(2^{n})} , but it is common to use faster heuristics to choose splits, such as k-means.